FINAL INDIVIDUAL PROJECT

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Analysis of the Relationship Between Physical Activity Frequency and BMI in Adolescents

1. RESEARCH OBJECTIVE

Investigate whether different levels of physical activity significantly affect adolescent BMI using the CDC YRBS 2007 dataset.

2. DATASET

- **Source:** CDC Youth Risk Behavior Survey (YRBS 2007)
- **Original Records:** 14,041
- **Valid Samples:** 12,894

3. DATA PROCESSING

- Removed missing values
- Calculated BMI from height and weight
- Recoded activity frequency into three groups

Low (0–2 days)

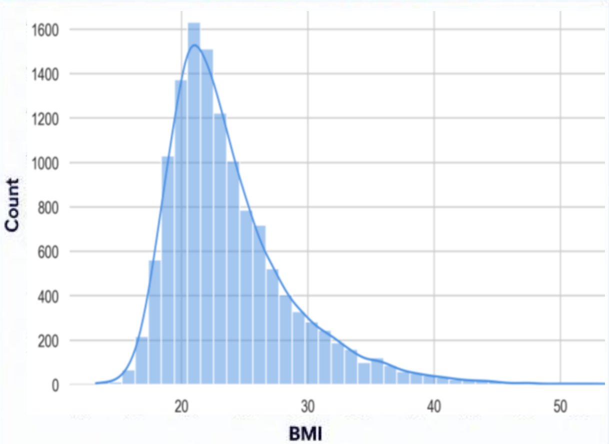
Moderate (3–4 days)

High (5–7 days)

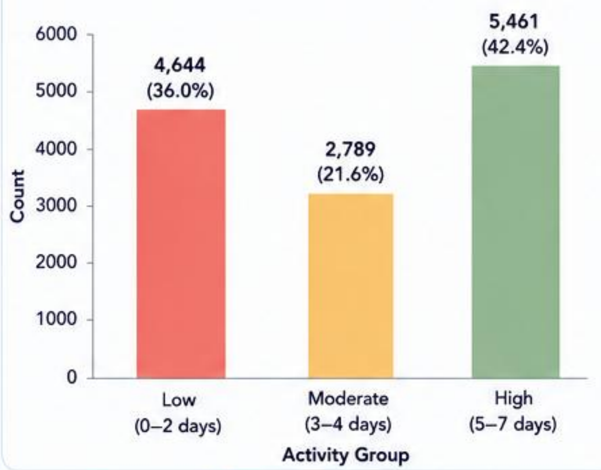
4. STATISTICAL METHODS

- Exploratory Data Analysis (EDA)
- One-Way ANOVA
- Tukey's HSD Post-hoc Test

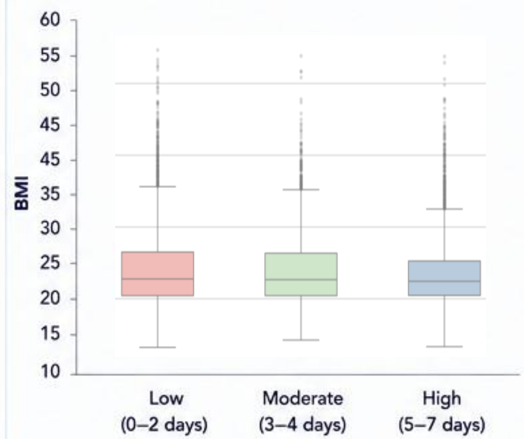
A. BMI DISTRIBUTION (HISTOGRAM)



B. PHYSICAL ACTIVITY GROUP DISTRIBUTION (BAR CHART)



C. BMI BY ACTIVITY GROUP (BOXPLOT)



KEY INSIGHT

The median BMI decreases as physical activity increases.

High activity group shows the lowest median BMI.

D. DESCRIPTIVE STATISTICS (BMI)

Activity Group	N	Mean	Std. Dev.	Median
Low (0–2 days)	4,644	24.12	5.39	22.76
Moderate (3–4 days)	2,789	23.91	5.05	22.66
High (5–7 days)	5,461	23.45	4.56	22.43

Mean BMI is lowest in the High activity group and highest in the Low activity group.

E. ANOVA RESULT

One-Way ANOVA

F-statistic	p-value
23.24	8.44×10^{-11}

$(p < 0.05)$

Conclusion: Significant differences exist among the three activity groups.

F. TUKEY HSD POST-HOC TEST RESULTS

Comparison	Mean Diff.	p-adj	Reject H_0	Result
High (5–7 days) vs Low (0–2 days)	0.664	0.0000	✓	Significant
High (5–7 days) vs Moderate (3–4 days)	0.451	0.0003	✓	Significant
Low (0–2 days) vs Moderate (3–4 days)	-0.213	0.1745	✗	Not Significant

* Negative mean difference indicates the first group has lower BMI.

G. FINAL CONCLUSION

- Higher physical activity frequency is associated with lower BMI among adolescents.
- High activity (5–7 days per week) results in significantly lower BMI compared to both Moderate and Low groups.
- Moderate (3–4 days) vs Low (0–2 days) shows no significant difference.

Regular exercise (5–7 days per week) provides the greatest health benefit.

PUBLIC HEALTH IMPLICATION: Encouraging adolescents to engage in physical activity for 5–7 days per week can effectively reduce obesity risk and promote long-term health.

Data Source: CDC YRBS 2007